Maximum Likelihood Estimation (MLE)

Complete Exam Answer Guide

11	4	7
Questions	Key Derivations	MLE Properties

Based strictly on IGNOU MTE-04 Unit 10 & Stanford CS109 Lecture Notes

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Q1 | 5 Marks | 2021-22 Final

What is point estimation? Define with an example: (1) Estimator (2) Estimate

Point Estimation

When we find a single value with the help of sample observations which is taken as the estimated value of an unknown population parameter, this value is known as a point estimate and the technique of estimating the unknown parameter with a single value is known as **point estimation**.

Example: If we estimate the average weight of men in a colony on the basis of the sample mean as 62 kg, then 62 kg is the point estimate of the average weight of all men in the colony.

(1) Estimator

An estimator is a rule or formula (a statistic) used to estimate the unknown population parameter. It is a function of the sample observations — not a single number, but the rule that produces one.

Example: x-bar = (1/n) * Sum(Xi) is an estimator of the population mean mu. It is the formula itself, applicable to any sample.

(2) Estimate

An estimate is the specific numerical value the estimator produces when applied to a particular sample. It is one realised number.

Example: After applying the formula to our sample, if we get 62 kg
estimate. The estimator is the formula; the estimate is its result.

Exam tip: Estimator = the formula/rule. Estimate = the actual number produced. Always state this distinction clearly in your answer.

Q2 | 5 Marks | 2021-22 Re-exam

What is Interval Estimation? Define: (1) Null Hypothesis (2) Alternate Hypothesis (3) Simple Hypothesis (4) Composite Hypothesis

Interval Estimation

Instead of finding a single value to estimate the unknown parameter, if we find two values between which the parameter may be considered to lie with a certain probability (confidence), this is known as an **interval estimate** and the technique is called **interval estimation**.

Confidence Interval = Sample Mean +/- Margin of Error

C% of such intervals will capture the true population mean mu

Example: If we estimate the average weight of men in an interval [40, 110] with 90% confidence that the actual average weight of all men lies in this interval, then [40, 110] is the interval estimate.

(1) Null Hypothesis (H0)

The null hypothesis is the statement that is presumed to be true. It always contains equality (=, >=, <=). It has no burden of proof.

Example: H0: mu = 200 (production rate has not changed)

(2) Alternate Hypothesis (H1)

The alternate hypothesis carries the burden of proof. It contains strict inequality (not=, >, <). Accepted only when H0 is rejected.

Example: H1: mu != 200 (production rate has changed)

(3) Simple Hypothesis

A hypothesis that completely specifies the distribution — it specifies the exact value of ALL parameters.

Example: H0: mu = 50 AND sigma-sq = 4 (bo

(4) Composite Hypothesis

A hypothesis that does NOT completely specify the distribution — at least one parameter is given as a range rather than a single value.

Example: H0: mu >= 50 (parameter given as range, not single val

Q3 | 5-10 Marks | 2021-22 Final & Re-exam | REPEATED

Define steps performed in MLE with proper statistical justification. Mention Properties of MLE.

What is MLE?

Maximum Likelihood Estimation (MLE) is one of the most important methods of point estimation. The principle of MLE is to find / estimate / choose the value of the unknown parameter which would **most likely generate the observed data**. The initial concept was given by Prof. C.F. Gauss but it was used as a general method by **Prof. Ronald A. Fisher in 1912**.

theta-hat = arg max L(theta) = arg max LL(theta)

theta-hat is the MLE estimate; L is likelihood; LL is log-likelihood

4 Steps of MLE — With Statistical Justification

1 **Write the Likelihood Function L(theta)**

Since data X1, X2, ..., Xn is IID (independent and identically distributed), the joint probability is the product of individual probabilities. For discrete: L(theta) = Product of P[X=xi]. For continuous: L(theta) = Product of f(xi, theta). Justification: the independence of observations allows multiplication of individual probabilities.

2 **Take the Log — Get LL(theta) = log L(theta)**

LL(theta) = Sum of log f(Xi | theta). Justification: log is a monotonically increasing function, so the arg max of L(theta) is IDENTICAL to the arg max of log L(theta). The product becomes a sum, which is far easier to differentiate. This is the key mathematical trick that makes MLE tractable.

3 **Differentiate and Set Equal to Zero**

Compute d(LL)/d(theta) = 0. This gives the likelihood equation. Solve it for theta to get the MLE estimate theta-hat. Justification: A necessary condition for a maximum of any differentiable function is that its first derivative equals zero at that point.

4 **Verify it is a Maximum (Not a Minimum)**

Check that d2(LL)/d(theta)2 < 0 at theta = theta-hat. Justification: A negative second derivative confirms the stationary point is a maximum. For two parameters (e.g. Normal with mu and sigma-sq), the Hessian matrix of second-order partial derivatives must be negative semi-definite.

7 Properties of MLE Estimators

1 **Asymptotically Unbiased**

Not necessarily unbiased for small samples. But as n goes to infinity, E(theta-hat) goes to theta. The bias disappears as sample size grows. Example: if E(theta-hat) = theta + 1/n, then as n increases, the term 1/n vanishes.

2 **Asymptotically Consistent**

For any epsilon > 0: P(|theta-hat - theta| < epsilon) goes to 1 as n goes to infinity. In plain terms: with enough data, the estimate converges to the true value in probability.

3 **Asymptotically Efficient**

When it exists, the MLE has the smallest variance among all consistent estimators in the large sample limit. It achieves the Cramer-Rao lower bound asymptotically.

4 **Linked to Sufficient Statistics**

If a sufficient statistic for theta exists for the distribution, the MLE will always be a function of that sufficient statistic.

5 **Invariance Property**

If theta-hat is the MLE of theta and gamma(theta) is any one-to-one function of theta, then gamma(theta-hat) is the MLE of gamma(theta). Example: for exponential distribution, theta-hat = 1/x-bar, so the MLE of e^theta is e^(1/x-bar).

6 **Not Necessarily Unique**

Multiple values of theta may maximise the likelihood equally. Uniqueness of the MLE is not guaranteed.

7 **Asymptotically Normally Distributed**

Under regularity conditions, the MLE follows a normal distribution asymptotically. This makes it easy to construct confidence intervals in large samples.

Q4 | 10 Marks | 2021-22 Re-exam

What is MLE? Generalise the steps involved in getting the MLE of an estimate. Mention Properties of MLE.

Definition of MLE

MLE is a method that determines parameter values in such a way that they maximise the likelihood that the process described by the model produced the data that were actually observed. It is one of the most important methods of point estimation because it generally gives very good estimators which possess most of the important properties of a good estimator.

Likelihood vs Probability — The Key Distinction

The likelihood function and the joint density function have the same functional form, but with opposite interpretations:

Joint density f(x1,...,xn | theta)	**Likelihood L(theta | x1,...,xn)**
Parameters theta are FIXED and KNOWN. The data is the variable. Answers: what data is probable given theta?	Data is FIXED and OBSERVED. Parameters theta are the variable. Answers: which theta is most plausible given this data?

L(theta) = Product[f(Xi | theta)] for i = 1 to n

Likelihood function — product of PMF/PDF across all n data points

LL(theta) = log L(theta) = Sum[log f(Xi | theta)]

Log-likelihood — product becomes sum; same arg max by monotonicity of log

The generalised steps and the 7 properties are identical to Q3 above. For a 10-mark answer: write all 4 steps with full statistical justification (as in Q3), then write all 7 properties with one example for the invariance property.

Special case: When calculus fails (parameter in the support, e.g. Uniform distribution), use the basic MLE principle directly. For Uniform(0, theta): theta-hat = x_(n) = largest observation. This is because L(theta) = (1/theta)^n is maximised when theta is minimised, and the minimum possible value consistent with all data points is the maximum observation.

Q5 | 10 Marks | 2021-22 Final & Re-exam — REPEATED VERBATIM | REPEATED

Define Unbiasedness. Test whether the following are unbiased estimators of mu: T1 = (X1+X2)/2 + X3 T2 = (X1+X2+X3)/3 T3 = (X1+3X2)/9 + 5X3 T4 = X1 - X2 + X3

Definition of Unbiasedness

An estimator T is said to be an **unbiased estimator** of the population parameter theta if the expected value of T equals theta for all values of theta:

E(T) = theta for all theta

If E(T) != theta, T is biased. The bias = E(T) - theta.

Assumption: X1, X2, X3 are IID random variables with E(Xi) = mu for all i. We test whether each estimator has expected value = mu.

Testing Each Estimator

T1 = (X1 + X2)/2 + X3

E(T1)	= E[(X1+X2)/2] + E(X3) = (1/2)mu + (1/2)mu + mu = 2mu

E(T1) = 2mu which is NOT equal to mu --> T1 is BIASED (bias = mu)

T2 = (X1 + X2 + X3) / 3

E(T2)	= (1/3)E(X1) + (1/3)E(X2) + (1/3)E(X3) = (1/3)mu + (1/3)mu + (1/3)mu = mu

E(T2) = mu --> T2 is UNBIASED (This is simply the sample mean x-bar)

T3 = (X1 + 3X2)/9 + 5X3

E(T3)	= (1/9)E(X1) + (3/9)E(X2) + 5*E(X3) = (1/9)mu + (3/9)mu + 5mu = mu*(1/9 + 3/9 + 45/9) = (49/9)mu

E(T3) = (49/9)mu which is NOT equal to mu --> T3 is BIASED

T4 = X1 - X2 + X3

E(T4)	= E(X1) - E(X2) + E(X3) = mu - mu + mu = mu

E(T4) = mu --> T4 is UNBIASED

Summary Table

Estimator	E(T)	Unbiased?
T1 = (X1+X2)/2 + X3	2mu	**NO — biased by mu**
T2 = (X1+X2+X3)/3	mu	**YES**
T3 = (X1+3X2)/9 + 5X3	(49/9)mu	**NO — biased by (40/9)mu**
T4 = X1 - X2 + X3	mu	**YES**

Quick check rule: Sum all the coefficients of X1, X2, X3 in the estimator. If they add up to 1, the estimator is unbiased. T2: 1/3 + 1/3 + 1/3 = 1 (Unbiased) | T4: 1 - 1 + 1 = 1 (Unbiased) | T1: 1/2 + 1/2 + 1 = 2 (Biased) | T3: 1/9 + 3/9 + 5 = 49/9 (Biased)

Q6 | 10 Marks | 2021-22 Final

Explain Efficiency. Define Relative Efficiency and mention all 3 conditions. Explain Consistency in detail and mention its properties.

Efficiency

Among all unbiased estimators of a parameter, the one with the **smallest variance** is called the most efficient estimator (or Minimum Variance Unbiased Estimator — MVUE). An estimator T1 is said to be more efficient than T2 if both are unbiased and Var(T1) < Var(T2).

Relative Efficiency

The relative efficiency of T1 with respect to T2 is:

$$e(T1, T2) = Var(T2) / Var(T1)$$

Ratio of variances — higher ratio means T1 extracts more information

Three conditions for relative efficiency:

Condition 1	If e(T1, T2) > 1, then T1 is MORE efficient than T2. Var(T1) < Var(T2).
Condition 2	If e(T1, T2) = 1, then T1 and T2 are EQUALLY efficient. Var(T1) = Var(T2).
Condition 3	If e(T1, T2) < 1, then T2 is MORE efficient than T1. Var(T2) < Var(T1).

Consistency

An estimator T is said to be a **consistent estimator** of theta if it converges in probability to theta as the sample size n goes to infinity. Formally, for any epsilon > 0:

$$\lim P(|T_n - \theta| < \epsilon) = 1 \text{ as } n \text{ goes to infinity}$$

In plain terms: with enough data, the estimator gets arbitrarily close to the true value

Properties of consistency:

• If T is an unbiased estimator of theta AND Var(T) goes to 0 as n goes to infinity, then T is consistent.

- The sample mean x-bar is a consistent estimator of mu (follows from the Law of Large Numbers).
- The sample variance s-sq is a consistent estimator of sigma-sq.
- If T is consistent for theta, then g(T) is consistent for g(theta) for any continuous function g.
- All MLE estimators are asymptotically consistent under standard regularity conditions.

Q7 | 10 Marks | 2021-22 Re-exam

Explain Sufficiency and Efficiency. Prove that 2x-bar is an unbiased estimate for theta when sampling from Uniform(0, theta).

Sufficiency

A statistic T(X1,...,Xn) is called a sufficient statistic for theta if the conditional distribution of the sample GIVEN T does not depend on theta. T captures all the information the sample contains about theta.

From MLE notes: if a sufficient statistic exists, the MLE will be a function of it (Property 4 of MLE).

Efficiency (recap)

Among all unbiased estimators, the most efficient has minimum variance. The MLE estimator is asymptotically the most efficient (Property 3 of MLE). An estimator is fully efficient if it achieves the Cramer-Rao lower bound.

e(T1,T2) = Var(T2)/Var(T1). If > 1: T1 more efficient. If = 1: equall efficient.

Proof: 2x-bar is Unbiased for theta (Uniform(0,theta))

Let X1, X2, ..., Xn be a random sample from Uniform(0, theta). The PDF is f(x, theta) = 1/theta for 0 <= x <= theta.

Step 1: Find E(X) for Uniform(0, theta)

E(X)	= Integral from 0 to theta of [x * (1/theta)] dx = (1/theta) * [x^2 / 2] from 0 to theta = (1/theta) * (theta^2 / 2) = theta / 2

Step 2: Find E(x-bar)

E(x-bar)	= E[(1/n) * Sum(Xi)] = (1/n) * n * E(X) = E(X) = theta/2

Step 3: Find E(2 * x-bar)

*E(2 * x-bar)*	= 2 * E(x-bar) = 2 * (theta/2) = theta

E(2 * x-bar) = theta --> Therefore 2x-bar is an UNBIASED estimator of theta. Proved.

Note: The MLE of theta for Uniform(0,theta) is theta-hat = x_(n) (maximum observation), which is actually BIASED. But 2x-bar is an alternative unbiased estimator. Both estimate the same theta.

Q8 | 5 Marks | 2021-22 Final

Explain the difference between Point Estimation and Interval Estimation.

Point Estimation vs Interval Estimation

Aspect	Point Estimation	Interval Estimation
Output	A single number	A range (lower, upper bound)
Example	mu-hat = 62 kg	mu in [58, 66] with 95% confidence
Certainty	No measure of accuracy given	Comes with a confidence level C%
Use case	Quick single-value estimate needed	When accuracy and reliability matter
Methods	MLE, method of moments, least squares	Confidence intervals using Z or t-dist.
Precision	No information about precision	Width of interval shows precision

Key quote from IGNOU notes: 'If we estimate the average weight as 62 kg — that is point estimation. If we estimate it as lying in [40, 110] with 90% confidence — that is interval estimation.' Both estimate the same parameter, but interval estimation provides additional information about precision.

C% of confidence intervals constructed this way will capture the true population mean mu

The confidence level C% is the overall capture rate if the method is used many times

Q9 | 10 Marks | 2021-22 Final | 2023-24 Re-exam | 2024-25 Final — VERY REPEATED | REPEATED

Derive the mathematical expressions for estimating the parameters of a Binomial distribution using MLE. Prove p-hat = x-bar / n.

Setup

Let X1, X2, ..., Xm represent the number of successes in m lots. Each Xi ~ Binomial(n, p) where n is KNOWN and p is UNKNOWN. We apply MLE to find p-hat.

P[X = x] = nCx * p^x * (1-p)^(n-x) for x = 0, 1, 2, ..., n

PMF of Binomial distribution

Step 1 — Write the Likelihood Function L(p)

L(p)	= P[X=x1] * P[X=x2] * ... * P[X=xm]
	= Product of [nCxi * p^xi * (1-p)^(n-xi)] for i=1 to m *IID assumption allows multiplication of individual PMFs*
	= [Product of nCxi] * p^(Sum xi) * (1-p)^(mn - Sum xi) *Collecting like terms: all p terms grouped, all (1-p) terms grouped*

Step 2 — Take the Log-Likelihood LL(p)

LL(p)	= log[Product nCxi] + (Sum xi)*log(p) + (mn - Sum xi)*log(1-p) *The first term is a CONSTANT with respect to p — it vanishes when differentiating*

Step 3 — Differentiate and Set Equal to Zero

d(LL)/dp	= (Sum xi)/p - (mn - Sum xi)/(1-p) = 0
Rearranging	(Sum xi)(1-p) = (mn - Sum xi) * p Sum xi - p*Sum xi = mn*p - p*Sum xi
Simplifying	Sum xi = mn * p
Solve for p-hat	p-hat = Sum(xi) / (mn) = x-bar / n *Since x-bar = (1/m) * Sum(xi), therefore Sum(xi)/mn = x-bar/n*

Step 4 — Verify it is a Maximum

d2(LL)/dp2	= -(Sum xi)/p^2 - (mn - Sum xi)/(1-p)^2
At p = p-hat	= -(m*x-bar)/p^2 - (mn - m*x-bar)/(1-p)^2 < 0 *Both terms are negative, so the second derivative is negative — confirmed MAXIMUM*

RESULT PROVED: p-hat = x-bar / n The MLE of p in the Binomial distribution equals the sample mean divided by n (number of trials). This result appears in 3 separate exam years — it is the most important MLE derivation to memorise.

Q10 | 5-10 Marks | 2023-24 Final & Re-exam | REPEATED

Explain how mathematically the mean and standard deviation of the Normal distribution are calculated through Maximum Likelihood Estimation.

Setup

Let X1, X2, ..., Xn be IID from N(mu, sigma-sq). We have TWO parameters to estimate: theta1 = mu (mean) and theta2 = sigma-sq (variance). The PDF is:

f(x, mu, sigma-sq) = (1 / sqrt(2*pi*sigma-sq)) * exp[-(x-mu)^2 / (2*sigma-sq)]

PDF of Normal distribution

Step 1 — Write the Likelihood Function

L(mu, sigma-sq)	= Product of (1/sqrt(2*pi*sigma-sq)) * exp[-(Xi-mu)^2/(2*sigma-sq)]
	= (1/(2*pi*sigma-sq))^(n/2) * exp[-(1/(2*sigma-sq)) * Sum(Xi-mu)^2]

Step 2 — Log-Likelihood

LL(mu, sigma-sq)	= -(n/2)*log(2*pi) - (n/2)*log(sigma-sq) - (1/(2*sigma-sq))*Sum(Xi-mu)^2

Step 3a — Estimate mu: Differentiate w.r.t. mu

d(LL)/d(mu)	= (1/sigma-sq) * Sum(Xi - mu) = 0 *Setting equal to zero*
Simplify	Sum(Xi) - n*mu = 0 --> mu-hat = (1/n) * Sum(Xi) = x-bar

mu-hat = x-bar (the sample mean)

Step 3b — Estimate sigma-sq: Differentiate w.r.t. sigma-sq

d(LL)/d(sigma-sq)	= -n/(2*sigma-sq) + (1/(2*sigma-sq)^2) * Sum(Xi-mu)^2 = 0
*Multiply by 2*sigma-sq^2*	-n*sigma-sq + Sum(Xi - mu)^2 = 0

Solve	sigma-sq-hat = (1/n) * Sum(Xi - mu-hat)^2 = (1/n) * Sum(Xi - x-bar)^2

sigma-sq-hat = (1/n) * Sum(Xi - x-bar)^2 --> sigma-hat = sqrt of that

Step 4 — Verify Both are Maxima (Hessian Check)

d2(LL)/d(mu)2	= -n/sigma-sq < 0 at mu = x-bar (Confirmed maximum for mu)
d2(LL)/d(sigma-sq)2	= -n/(2*sigma-sq^2) < 0 at sigma-sq = s'^2 (Confirmed maximum for sigma-sq)
Hessian determinant	= n^2 / (2*sigma-sq^3) > 0 (Joint maximum confirmed)

All three conditions of the Hessian are satisfied. The likelihood function is MAXIMUM at mu = x-bar and sigma-sq = s'^2. Proved. Important note: The MLE divides by n (not n-1). The usual sample variance s^2 = Sum(Xi-x-bar)^2/(n-1) is unbiased. The MLE sigma-sq-hat = Sum(Xi-x-bar)^2/n is slightly biased but asymptotically unbiased (bias vanishes as n increases).

Q11 | 5 Marks | 2022-23 Final

How to estimate the likelihood function for Binomial distribution for parameter estimation? (Explain with formula)

The Likelihood Function for Binomial Distribution

Let X1, X2, ..., Xm be a random sample from Binomial(n, p) where n is known. The PMF is:

P[X = x] = nCx * p^x * (1-p)^(n-x)

PMF of Binomial distribution

Since Binomial is a discrete distribution, the likelihood function is the joint probability — the product of individual PMFs across all m observations:

L(p) = Product[P[X = xi]] for i = 1 to m

Joint probability of all observations under independence (IID assumption)

L(p) = Product[nCxi * p^xi * (1-p)^(n-xi)]

Substituting the Binomial PMF for each observation

L(p) = [Product nCxi] * p^(Sum xi) * (1-p)^(mn - Sum xi)

Collecting like terms: all p's grouped, all (1-p)'s grouped

The Log-Likelihood

Taking the natural log converts the product to a sum:

LL(p) = log[Product nCxi] + (Sum xi)*log(p) + (mn - Sum xi)*log(1-p)

The first term is a constant w.r.t. p and drops out when differentiating

The likelihood function answers: for which value of p would this exact observed sample have been most probable? Setting d(LL)/dp = 0 and solving gives p-hat = x-bar/n (full derivation in

Q9 above).

> The likelihood function L(p) = [Product nCxi] * p^(Sum xi) * (1-p)^(mn - Sum xi) is a function of p given fixed data. It is maximised at p-hat = x-bar/n — the proportion of successes per trial across the sample.